

Institute of Actuaries of India

Subject ST7 – General Insurance: Reserving and Capital Modeling

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Amounts in Rs crores**

Sr.	Particulars	2013	2014	2015	2016	2017	2018
1	Earned Premium for each year	8	24	64	152	240	264
2	Claims Paid in each year	3	12	40	103	186	196
3	Reported Claims O/s at start of each year	0	2	9	17	35	49
4	Reported Claims O/s at end of each year	2	9	17	35	49	74
5	IBNR at start of each year	0	2	4	12	24	34
6	IBNR at end of each year	2	4	12	24	34	44
7	Commission for each year	1	2	4	8	12	13
8	Expenses for each year	2	4	8	16	24	26
9	Underwriting Result for each year (9 = 1+2-3+4-5+6+7+8)	-2	-3	-4	-5	-6	-6

[2]

ii) Amounts in Rs crores

Sr.	Particulars	2013	2014	2015	2016	2017	2018
1	Written Premium for each year	25	50	100	200	300	325
2	Earned Premium for each year	8	24	64	152	240	264
3	Claims Paid in each year	3	12	40	103	186	196
4	Reported Claims O/s at start of each year	0	2	9	17	35	49
5	Reported Claims O/s at end of each year	2	9	17	35	49	74
6	IBNR at start of each year	0	2	4	12	24	34
7	IBNR at end of each year	2	4	12	24	34	44
8	Incurring Claims for each year (8 = 3-4+5-6+7)	7	21	56	133	210	231
9	Incurring Claims Ratio for each year(9 = 8/2)	87.5%	87.5%	87.5%	87.5%	87.5%	87.5%
10	Commission for each year	1	2	4	8	12	13
11	Expenses for each year	2	4	8	16	24	26
12	Commission Ratio for each year (12=10/1)	4.0%	4.0%	4.0%	4.0%	4.0%	4.0%
13	Expense Ratio for each year (13 = 11/1)	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%
14	Combined Ratio for each year (14=9+12+13)	99.5%	99.5%	99.5%	99.5%	99.5%	99.5%

[4]

iii) The underwriting losses for EUCL are increasing year on year from -2 to -6...

...whereas the combined ratio for UECL is constant at 99.5% for all the years

This anomaly is due to different denominators used for combined ratio ...

...also predominantly due to effect of growing business and no allowance for Deferred Acquisition Cost in underwriting result

This results only in timing difference in recognition of underwriting profits...

.. as the profits get locked into unearned premium reserves as the commissions are not deferred.

UECL is overall profitable on underwriting basis since combined ratio is below 100%...

...also if investment income is added; the overall profits of the company is higher.

[5]

iv) Cumulative Paid Claims Triangle

Accident Year/Lag	0	1	2	3	4	5
2013	3	5	7	8	8	8
2014	10	14	18	22	25	
2015	34	44	52	59		
2016	88	107	121			
2017	155	170				
2018	157					

Incremental Incurred Claims Triangle

Accident Year/Lag	0	1	2	3	4	5
2013	5	1	1	0	1	0
2014	18	2	2	1	2	
2015	45	5	5	7		
2016	114	5	8			
2017	188	3				
2018	201					

[4]

v) Assumptions–

1. The latest year is fully developed by 5-years delay from accident year.
2. Claims Inflation is implicitly modelled in the triangulation

Paid Claims Projection

Cumulative Paid Claims Triangle

Accident Year/Lag	0	1	2	3	4	5
2013	3	5	7	8	8	8
2014	10	14	18	22	25	
2015	34	44	52	59		
2016	88	107	121			
2017	155	170				
2018	157					

Age to Age Factor	1.172	1.165	1.156	1.100	1.000	1.000
Age to Ultimate Factor	1.736	1.481	1.271	1.100	1.000	1.000

Accident Year	2013	2014	2015	2016	2017	2018	Total
Ultimate Claims	8.00	25.00	64.90	153.84	251.74	272.58	776.06
Less Paid Claims	8.00	25.00	59.00	121.00	170.00	157.00	540.00
Less Reported O/S	-	-	3.00	6.00	21.00	44.00	74.00
IBNR Claims	-	-	2.90	26.84	60.74	71.58	162.06

Incurred Claims Projection

Cumulative Incurred Claims Triangle

Accident Year/Lag	0	1	2	3	4	5
2013	5	6	7	7	8	8
2014	18	20	22	23	25	
2015	45	50	55	62		
2016	114	119	127			
2017	188	191				
2018	201					

Age to Age Factor	1.043	1.082	1.095	1.100	1.000	1.000
Age to Ultimate Factor	1.360	1.304	1.205	1.100	1.000	1.000

Accident Year	2013	2014	2015	2016	2017	2018	Total
Ultimate Claims	8.00	25.00	68.20	153.00	248.99	273.36	776.55
Less Incurred Claims	8.00	25.00	62.00	127.00	191.00	201.00	614.00
IBNR Claims	-	-	2.90	26.84	60.74	71.58	162.55

[7]

vi)

Particulars	2013	2014	2015	2016	2017	2018
Financial Incurred Claims ratio with booked IBNR	87.5%	87.5%	87.5%	87.5%	87.5%	87.5%
Financial Incurred Claims ratio without IBNR	62.5%	79.2%	75.0%	79.6%	83.3%	83.7%

The underwriter is referring to the financial incurred claims ratio. From this point of view, what the underwriter says is true that these ratios have consistently been below the assumed expected ULR of 87.5%.

However, this approach is inappropriate since in a rapidly growing book the financial incurred claims ratio does not correspond/match with the earned exposure due to delay in claims reporting and settlement.

The portfolio may not be as short tailed as the underwriter believes. This is evident from the development factors calculated. For example age to age factor is as high as 1.1 for year 3 to 4

Also while claims reporting is shorter tailed, there are considerable delays in settlement of claims. For example paid development factors are greater than incurred development factors.

Particulars	2013	2014	2015	2016	2017	2018
Accident year ULR as per incurred CLM method	100.0%	104.2%	106.6%	100.7%	103.7%	103.5%
Accident year Loss ratio without IBNR	100.0%	104.2%	96.9%	83.6%	79.6%	76.1%

Even if the underwriter's judgment is to be used for ULR method, then it should be on accident year cohort or on underwriting year cohort. [5]

vii) The IBNR currently held in the books as at end of 2018 = Rs 44 cr

IBNR as calculated by Paid CLM method = Rs 162.06 cr

IBNR as calculated by Incurred CLM method = Rs 162.55 cr

The IBNR currently held in books is significantly less than actually required by Chain Ladder methods.

The revised combined ratio for UECL for 2018 is

103.5% (Incurred claims ratio as per incurred CLM) + 4% + 8% = 115.5%

This would result in considerable underwriting losses and overall losses even after investment income. The view on performance changes from (c) and after the analysis, it turns out that the company is unprofitable. [3]

viii) Assumption –

1. Selected incurred claims ULR for Accident year 2018 is based on incurred CLM since it is higher than paid CLM
2. The unearned exposure as at end of 2018 will be same as the incurred claims experience for accident year 2018.
3. Out of 8% expense ratio, 2% is applicable for claims related expense which will be used in URR calculation.

Unexpired Risk Reserve = Unearned Premium Reserve + Additional Unexpired Risk Reserve

= (Total Written Premium – Total Earned Premium) + Excess of future claims and claims related expenses payment over and above UPR

= (1000 – 752) + (103.5% + 2% - 100%) * (1000-752)

= 261.48

[2]

[32 Marks]

Solution 2:

- i) A clause within a catastrophe reinsurance treaty that specifies the limited period during which claims can be aggregated for the purpose of one claim on the reinsurance contract. Commonly 24 or 72 hours are used.

The purpose of the hour's clause is to prevent any disputes occurring in cases where, for example, two different storms occur four days apart (in which case the two storms are treated as separate events). The starting point will be chosen by the insurer in order to maximise the reinsurance recoveries.

[2]

- ii) A financial guarantee issued by a bank that permits the party to which it is issued to draw funds from the bank in the event of a valid unpaid claim against another party.

Lloyds allow capital by way of Letter of Credit. Some jurisdictions may allow it as a valid source of capital in their solvency regulation.

Letter of Credit are usually used in Fronting Agreements

Also Letter of Credit are common for export sales and are referred to in Marine Insurance [2]

- iii) The Ogden tables are actuarial tables that provide multipliers that are widely used by UK courts in assessing the present value of future losses in personal injury or accident death cases.

They are based on assumptions for life expectancies, employment risk and a set of discount rates.

The discount rate actually used is set by the Government. A low discount rate will result in a high lump sum award.

The discount rates to be used (and indeed other assumptions in the tables) will be changed from time to time. Some changes could have a dramatic effect upon the overall claim payments. The unpredictability of these court decisions makes it difficult to estimate this type of inflation. [2]

[6 Marks]

Solution 3:

1. Liability

- a. **Public Liability** – this will cover indemnity against legal liability for the death of or bodily injury or for damage to property belonging to visitors at the stadiums, participating athletes and any other third party due to negligence of FISA and/or involvement in sport activities.

Measure of Exposure – Number of Visitors or Events or Ticketing Revenue

- b. **Director's and Officer's Liability**—this will indemnify FISA against the legal liability to compensate third parties owing to any wrongful act of the FISA Officials e.g. incorrect decisions, wrong declaration of results, incorrect disqualification of athletes etc.

Measure of Exposure – Number of Officers/Employees or Events or Total Revenue

- c. **Environmental Liability**—this will indemnify FISA against the legal liability to compensate third parties as a result of bodily injury, death and damage to property as a result of unintentional pollution for which the FISA is deemed responsible.

Measure of Exposure – Number of Visitors or Events or Ticketing Revenue

- d. **Employers Liability** – this will cover legal liability to compensate an employee or his or her estate for bodily injury, disease or death suffered, owing to negligence of the employer, in the course of employment.

Measure of Exposure – Number of Employees or Payroll

- e. **Motor Third Party Liability** – this will cover legal liability to compensate a third party for bodily injury, disease or death suffered, owing to usage of its motor vehicles for transport of equipment and staff to different locations.

Measure of Exposure – Number of Vehicle-days or Transits

2. Property Damage

- a. **Moveable Property/ Sports Equipment** - This will cover any loss, damage or theft of moveable property or sports equipment of FISA. This will include trophies and other valuables owned by FISA. This can be for replacement value after depreciation or 'new for old' basis.

Measure of Exposure – Sum Insured Days or on First Loss Basis

- b. **Marine Insurance** – This will cover any loss or damage to sports equipment/other goods in transit to different locations.

Measure of Exposure – Consignment Value

- c. **Specialized installed equipment and/or any other electrical fittings** - This will cover any loss or damage to specialized installed equipment and/or any other electrical fittings of FISA. These can be for specified perils or on all-risk basis.

Measure of Exposure – Sum Insured Days or on First Loss Basis

- d. **Commercial Property** – This will cover any loss or damage to corporate buildings of FISA and its own stadiums. These can be for specified perils or on all-risk basis.

Measure of Exposure – Sum Insured Days or on First Loss Basis

- e. **Motor Insurance** - This will cover any loss or damage to motor vehicles of FISA used for transport of equipment and staff to different locations.

Measure of Exposure – Number of Vehicle days or Transits

3. Financial Loss Cover

- a. **Event Cancellation** - This will cover FISA for any financial losses faced due to cancellation of a sport event due to unforeseen circumstances like natural or manmade catastrophes like earthquake, acts of terrorism etc. It will help FISA to refund all the ticket revenues as well as to compensate for loss of revenue from global broadcasting.

Measure of Exposure – Total Revenue or Number of Event-days

- b. **Cyber/Broadcast Interruption Insurance**- This will cover FISA for any financial losses faced due to cyber-attacks on its official website and any interruption in global broadcasting.

Measure of Exposure – Total Revenue or Number of Countries of Broadcasting

- c. **Contingent Business Interruption** - This will cover FISA for any financial losses faced due to an unforeseen event at the premises of its suppliers of sports equipment. This could include last moment purchase of such sport equipment at higher costs from other suppliers.

Measure of Exposure – Number of Suppliers or Total Purchases

- d. **Fidelity Guarantee**-This will cover FISA for any financial losses caused by dishonest actions of its employees (fraud or embezzlement). These will include loss of money/goods and also reasonable fees incurred in establishing the size of the loss

Measure of Exposure – Number of Employees or Ticketing Revenue in Cash

4. Fixed Benefit Cover

- a. **Personal Accident** – This will provide a specified fixed benefit amount in the event that the employees of FISA/ participating athletes suffer loss of one or more limbs or other specified injury or accidental death.

Measure of Exposure – Number of Employees and Athletes

[12 Marks]

Solution 4:

i)

1. Future Written Premium Income

- a. Need to project by line of business (subject to size of each line, grouping of business may be required)
- b. Figures may also be split by source of business
- c. Assumptions
 - i. Following consultation with sales managers/underwriters
 - ii. Earned Premium could also be used depending on capital model structure/requirement

2. Future Claims split into

- a. Attritional/large/catastrophe
- b. IBNER
- c. IBNR
- d. Claims from the period of unexpired risk on existing business
- e. Claims from business that is written in the future

3. Future Expenses

- a. Some assumptions are likely easy as simply a function of other assumptions in the business plan. Example, commissions is a percentage of written premium
- b. Others such as staff costs or rentals will be more difficult to predict as they will depend on the projected business growth and staffing plan

4. Investment Return

- a. Assumptions
 - i. Current investment portfolio held
 - ii. Investment prospects and expectations about the future economic environment
 - iii. Current and projected future investment policy of general insurer

5. Ceded reinsurance
 - a. Assumptions
 - i. Any existing arrangements
 - ii. Any changes that could be made to those arrangements
 - iii. Possible new arrangements that could be put in place, particularly where there are forecast changes in underlying direct exposures
 - iv. Any expected softening or hardening of future reinsurance costs
6. Economic environment
 - a. Assumptions
 - i. Future inflation
 - ii. Future interest rates
7. Insurance Cycle
 - a. Assumptions
 - i. Different classes of business may be at different stages in the cycle
8. Risk Measure
 - a. Requires a defined risk measure on which it will be calibrated. Includes the type and confidence.

[15]

ii) Insurance risk

1. Lower than expected levels of new business or an unexpected mix of business eg smaller policies than expected, will result in lower premium income than expected.
2. Premium income is often used to pay current liabilities so that a more long-term investment strategy can be followed.
3. A reduction in premium income can lead to a lack of liquid assets with which to meet current liabilities. This is a type of underwriting risk.
4. Higher than expected claims can lead to liquidity issues if there are not enough liquid assets available to pay the claims. This is a type of reserving risk.
5. High claims may be due to a higher than expected level of attritional claims...
6. ...more large claims than expected
7. ...Catastrophe claims
8. ...emergence of latent claims

Credit Risk

1. Parties who may owe the insurer money may include reinsurers, owing reinsurance recoveries...
2. ...brokers, owing premium balances
3. ...other debtors
4. Each of these parties might delay the payments
5. Or default on the payments

6. Each of these events could lead to a liquidity risk if the insurer was depending on such payments to meet its current liabilities.

[7]

[22 Marks]**Solution 5:**

1. Completeness of data, reconciliation with accounting information
2. Checks of frequency and severity over time
3. Comparison with data used in previous review
4. If loss ratios are used, check attachment of claims to policy data
5. Check for possible corruption of data during cleaning process
6. Check data is correctly fed through reserving software
7. Check estimates against those used by pricing department
8. Check that frequency and severity combine correctly. Example by origin year, type of claim
9. Check inflation or other trends have been projected to expected claim payment date
10. Check allowance for reinsurance recoveries is consistent with reinsurance programs in place
11. Check that appropriate allowance has been made for individual large claims and accumulations
12. As consulting actuary, you are likely to have information regarding competitor's reserves for similar business and can check against this
13. Check appropriate allowance has been made for claims handling expenses or separate reserve has been set up for this
14. Check that recent and appropriate data is used
15. Check credibility of data in cells and homogeneity within cells
16. Check suitability of model and data cells
17. Check that any changes in the reserves look reasonable by comparing with previous reviews

[8 Marks]**Solution 6:**

1. Data Protection Laws
 - i. What data a company may hold on its customers
 - ii. Both personal and commercial
 - iii. For how long
 - iv. What a company may use its data for
 - v. Rights of access by a customer to its personal data
 - vi. Rights of a customer to have incorrect data corrected
 - vii. Parties authorized to handle data at data gathering stage
 - viii. ...at Data Processing stage
 - ix. ...subsequently
 - x. Guidelines for third parties (such as consultancies) to adhere to when dealing with data
 - xi. Requirements to destroy personal data after use
 - xii. Security requirements during data storage

xiii. ...during transfer to other parties.

2. Consequences of violation

- i. Criminal prosecution of responsible parties
- ii. Legal action against company
- iii. Fines to the company
- iv. Requirements to compensate the customer
- v. Restrictions on the future activities of the company
- vi. More frequent monitoring of the company
- vii. Negative publicity of the company (and the individuals) which would impact business volumes.

[10 Marks]

Solution 7: Capital will be needed to cover:

- 1. Product development costs, marketing the new product, admin systems
- 2. Administrative expenses of selling new business
- 3. Commission
- 4. Setting up of reserves, including solvency margin
- i. Necessary to create model points that will represent the mix of business that the company expects to sell.
- ii. A cashflow model will be needed that projects cashflows into the future. Includes premium, claims, expenses, investment return, reinsurance premiums, recoveries, tax
- iii. Model must allow for required solvency margins
- iv. Cashflows should be discounted (@ appropriate disc. Rate) in order to determine the reserves that are needed.
- v. If calculating economic capital, realistic basis might be appropriate
- vi. If calculating regulatory capital, prudent or cautious basis may be appropriate
- vii. Need to decide which variables will be modelled deterministically and which ones stochastically
- viii. Time horizon over which to model cashflows – should be lifetime of business being modelled
- ix. Degree of accuracy required
- x. Correlations between different variables should be allowed for
- xi. Model should ensure consistency between various assumptions
- xii. Allowance for guarantees on the contract
- xiii. Future tax position should also be considered
- xiv. Scenario and sensitivity testing should be carried out on the different assumptions
- xv. Output from the model will be the amount that needs to be held as reserves, plus any required solvency margin. This will be scaled up to reflect the volume of business that the company expects to sell.
- xvi. The capital required for product development will be added in to give the total capital required to set up the new line of business.

[10 Marks]
